Personal Statement

"A shipping route may appear as a clean line on a chart, yet every movement along it is shaped by weather, traffic, regulation, cost, environmental constraints, and human judgment." This contrast has guided my academic development in maritime affairs management. I began by studying individual questions—how to assess navigation risks, forecast port throughput, or identify vessel routes—but gradually realized that they are interconnected parts of a larger system. What makes a maritime system truly intelligent is not the sophistication of one model, but its ability to turn fragmented information into decisions that remain safe, efficient, resilient, and responsible under uncertainty.

My first systematic encounter with this complexity came during my internship at Guangzhou Xinhai Maritime Technology Services Co., Ltd. I participated in navigation safety assessments for drilling-platform operations, subsea pipeline and cable maintenance, vessel towing, and other activities affecting waterways. While examining natural conditions, traffic patterns, and operational requirements, I learned that no safety measure works in isolation. Warning signs, communication procedures, guard vessels, operational windows, and emergency arrangements must function as one coordinated system. Helping organize expert reviews and compile safety assurance plans taught me that maritime management is fundamentally a problem of decision-making under uncertainty, where technical evidence must be translated into clear and workable procedures.

I later encountered the social dimension of maritime development through the Nanfengchuang Research China project on offshore engineering and fishermen's livelihood transformation in Huilai County. Our team interviewed and surveyed fifty fishing households and analysed their income, quality of life, perceptions of offshore wind development, and willingness to pursue alternative livelihoods. Using KMO testing, factor analysis, principal component analysis, and K-means clustering, we identified the factors shaping their attitudes. Yet the fieldwork revealed something statistics alone could not: the same project may represent clean-energy progress to policymakers, restricted fishing space to communities, and uncertainty to individual families. I therefore came to understand sustainability as both an environmental and a social responsibility.

With this broader perspective, I began exploring how quantitative methods could support strategic decisions in ports. As leader of the Qinglan e-Plan project, I analysed Shenzhen Port's cargo throughput from 2013 to 2023. I developed a GM(1,1) grey prediction model, introduced Markov-chain corrections, and improved the state correction factors through nonlinear optimization. Using MSE and MAPE to evaluate performance, I produced forecasts for 2024–2026. The work resulted in my first-author publication and later supported further research on a CPSO-improved forecasting approach. More importantly, it trained me to complete an end-to-end research process: identifying a practical question, collecting and preprocessing data, constructing and validating a model, interpreting its limitations, and communicating results clearly.

My interest in maritime data then expanded from temporal forecasting to spatial analysis during my internship at Guangzhou Yihai Shipping Co., Ltd. Working with AIS vessel trajectory data, I supported route identification, traffic-density analysis, image processing, and offshore wind-farm site selection using Python and GIS. By cleaning and visualizing large-scale trajectories, I helped identify major traffic corridors and examine potential conflicts between established shipping patterns and proposed offshore developments. Missing records, irregular tracks, heterogeneous vessel types, and changing operating conditions also showed me the difficulty of converting raw data into reliable planning evidence. This experience strengthened my interest in maritime informatics and in integrating AIS data, geographic information, risk assessment, and optimization.

The Fifth International Supply Chain Modeling and Design Competition broadened my perspective from maritime transport to end-to-end logistics systems. As team leader, I addressed a low-carbon spare-parts distribution problem involving facility location, inventory allocation, service levels, and operational robustness. I standardized five business datasets with the Haversine algorithm, formulated set-covering and mixed-integer programming models, and used Gurobi to design a five-node distribution network. I then combined ABC-XYZ classification with service-level sensitivity analysis to develop differentiated safety-stock strategies and used SimPy-based discrete-event simulation to test order fulfilment and inventory stability. This project taught me that optimization does not end when a solver produces an objective value; a solution must also remain effective under operational variability.

This need to connect models with operations became clearer during my internship with Worldwide Logistics Group. My work includes issuing air waybills, checking commercial invoices and packing lists, supporting airline booking and capacity allocation, coordinating carriers and overseas agents, and verifying HS codes, trade terms, and requirements for general and dangerous goods. These tasks may appear procedural, yet international logistics depends on the accuracy of thousands of such procedures. A small inconsistency can interrupt an entire shipment. By helping improve document-checking and data-filing procedures, I gained a grounded understanding of information quality, compliance, and traceability. I learned that digital platforms and algorithms can improve logistics only when the underlying data and workflows are reliable.

Across these experiences, I have developed a research vision connecting maritime data, decision-making, and system governance. I hope to use forecasting, optimization, simulation, and spatial analysis to support port planning, route design, logistics management, and maritime risk control, while considering environmental objectives, operational uncertainty, regulation, and stakeholder needs. My academic path has moved from observing individual operations to understanding the wider networks connecting vessels, ports, logistics providers, coastal communities, and regulatory institutions. In advanced study, I hope to deepen my knowledge of maritime economics, transport analytics, intelligent shipping, and sustainable logistics. Ultimately, I want to contribute to maritime systems in which efficiency, safety, sustainability, and resilience are treated as interconnected dimensions of responsible decision-making.